Q2 2019 – Q1 2020 BIKE-SHARE REPORT

TO: LILY MONROE & CYCLISTIC EXECUTIVE TEAM
FROM: JACOB RODRIGUEZ
SUBJECT: DATA ANALYSIS FOR FUTURE MARKETING CAMPAIGN
DATE: 04/15/2020*

SUMMARY

Because of the increased lifetime value of annual members compared to casual riders, the aim of Cyclistic's next marketing campaign will have a specific focus on converting casual riders to annual members. In order to better understand the casual rider market segment, an analysis of trip data from the past year was performed to understand how behavior differs between casual and member riders. From this analysis, it was found that casual riders' behavior aligns with a focus on recreational use and their demographics lean younger and more female than annual members. It is the suggestion of this analysis that the next marketing campaign target casual riders, ages 16-34, and begin before the peak usage month in August.

CASUAL RIDER DEMOGRAPHICS

Casual riders account for 23.35% of all trips in Chicago. (see Fig. 1) Nearly 50% of trips taken by casual riders come from those between the ages 25-34 and 25.49% of riders ages 16-24. (see Fig. 2) Casual rider trips taken by females make up 38.3% of trips, compared to the 25.69% of trips taken by annual member female riders. (see Fig. 3) Overall, casual riders skew slightly younger than annual member riders who have approximately the same percentage of trips from ages 25-34 but only 9.67% of trips from ages 16-24. Overall, the number of trips taken in a month by casual riders seems to correlate with the number of trips taken by members, indicating similar seasonal behavior, although at different scales. (see Fig. 4)

BEHAVIOR

Casual riders' trips are significantly longer than those of annual members with the average trip for casual riders being 58.85 minutes, more than four times longer than members' average trip time of 14.16 minutes. (see Fig. 5) Casual riders' number of trips peak on Saturday and Sunday where their market share of trips is doubled compared to the weekdays. Casual riders account for approximately 40% of all trips taken on Saturday and Sunday when casual riders only account for 23.35% of all trips. (see Fig. 6)

STRATEGIC CONTEXT

Based on the findings from the past year and taking into consideration the semi-economic shutdown from COVID-19, the time to launch this campaign, circumstances permitting, would be late Spring, when ridership is expected to increase until its probable peak in Summer.

The target market for this campaign that best suits the objective of converting casual riders into annual members would be current riders, ages 16-34, in the Chicago area. The messaging of this campaign should be inclusive to both genders in the goal of increasing female ridership which is relatively low among current annual members. Additionally, securing this market maximizes lifetime customer value.

Usage patterns suggest that casual riders' use cases differ from annual riders by a difference in the days of the week they take the most trips and the average duration of their trips. The pattern of longer ride times and increased activity on Saturday aligns with recreational use.

COVID PANDEMIC

In March of this year, COVID-19 was declared a national emergency and Governor Pritzker announced a statewide stay-at-home order that took effect on March 21st and has been extended to April 30th. An extension past July would likely significantly reduce the outcomes of any campaign as ridership will already be close to its peak and expected to decline. In the event a full marketing campaign is not viable, a soft launch can be conducted to test marketing material. Continued analysis as well as conducting qualitative data collection and analysis can further assist the development of a campaign to be launched before the next peak season following the conclusion of the pandemic.

DATA PROCESS

Data used for analysis was obtained through a web index, uploaded into a Postgres database, and queried through DBeaver. Fields were renamed to keep fields consistent and then aggregated into a single set. The day of week field was calculated using the start time. For full documentation of data manipulation and validation, refer to this project's data log.

DATA LIMITATIONS

Data analyzed included data for Cyclistic rides taken between April 2019 – March 2020. Data did not include a user id key to find real population data; rather, it displays the count of rides in that category. This means that the 25-34 age demographic is not necessarily 50% of users, only that they take 50% of trips. Recommendations for addressing this gap are outlined in the following section.

Based on a single year of data, seasonal observations like a peak in August may not hold true for all years. Trip data only included gender and birth year for the first three quarters. Additionally, both fields were optional. Null data was filtered out for visualizations. Gender was missing from 24.91% of trips and birth year was missing from 24.44% of trips. Of the months that contained gender and birth year, only 15.62% were missing gender and 15.1% were missing birth year. At this time, there is no reason to assume the missing data is skewed - for example, that one gender was less likely to report than another.

ANALYSIS NEXT STEPS

The birth year and gender fields should be added to this dataset for the first three months of 2020 to ensure accuracy if available. Matching trip data to rider id can provide more actionable insights for the campaign such as revealing the population size of the target demographic. If available, location data should be added to Q2 2019 – Q4 2019 and an analysis done on geographic location to find any relevant hot spots. If relevant qualitative data is not available for the target market, it should be acquired. Surveys and interviews can assist in the development of campaign messaging and executions.

RECOMMENDATIONS

Accepting that increasing the number of annual members is key for Cyclistic's future success, casual riders offer a significant opportunity. Their needs and usage patterns are distinct from current annual members, and a well-timed, targeted campaign has the potential to convert a meaningful share of this segment. The behavioral and demographic profile identified in this analysis provides a foundation for that effort.

The following are advised:

1. Launch campaign in late Spring, targeting riders 16-34 with a focus on recreational ridership.
2. Design inclusive messaging to grow female annual membership.
3. If COVID restrictions extend past July, conduct a soft launch until a full campaign is viable.

DATA VISUALIZATIONS

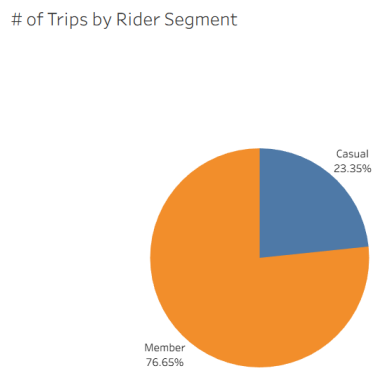

Figure 1: # of Trips by Rider Segment- Casual riders make up 23.35% of trips.

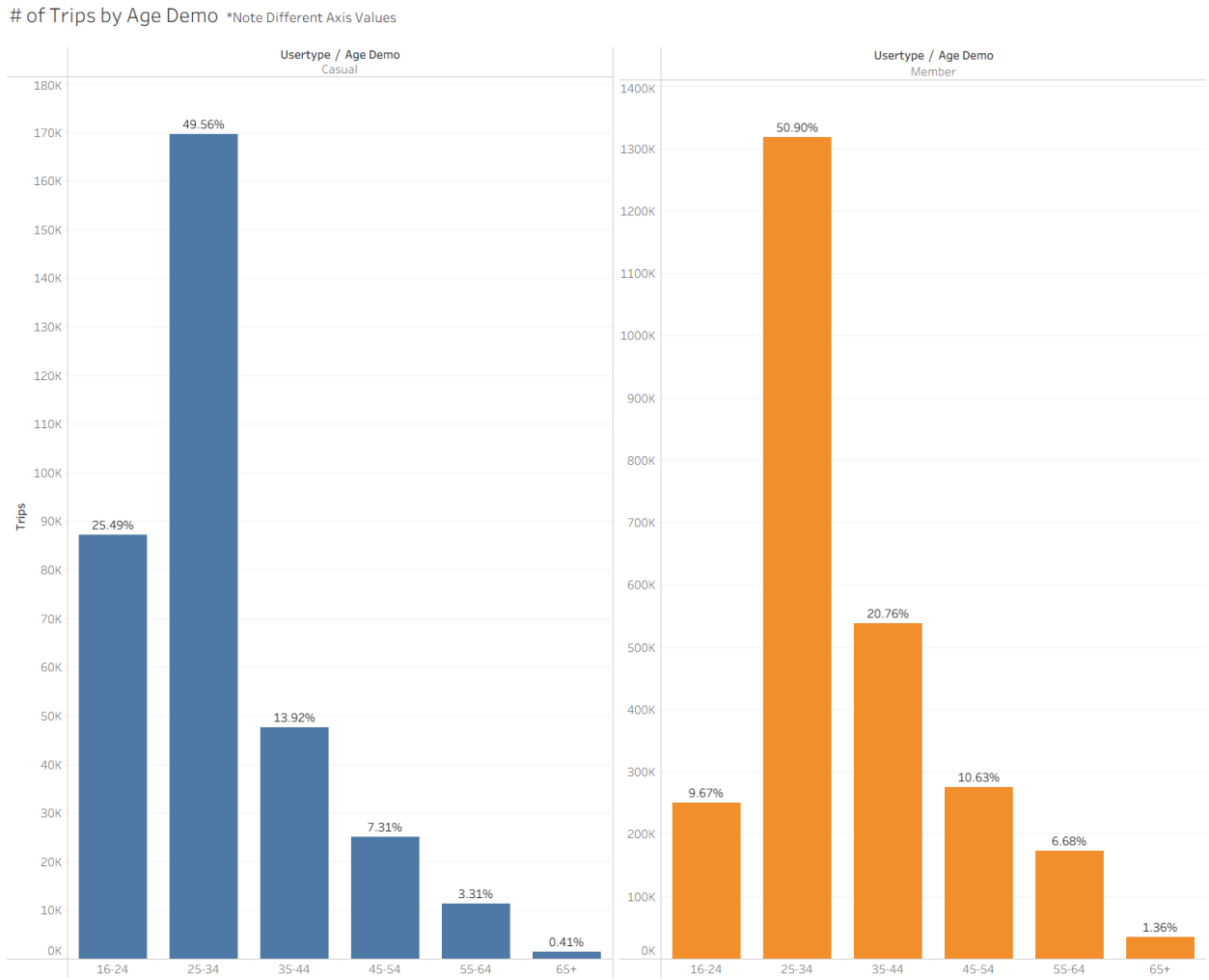

Figure 2: # of Trips by Age Demo- Rides are concentrated in the 25-34 age demo between both casual riders and annual members. 16-24 is considerably large for casual riders at 25.49%.

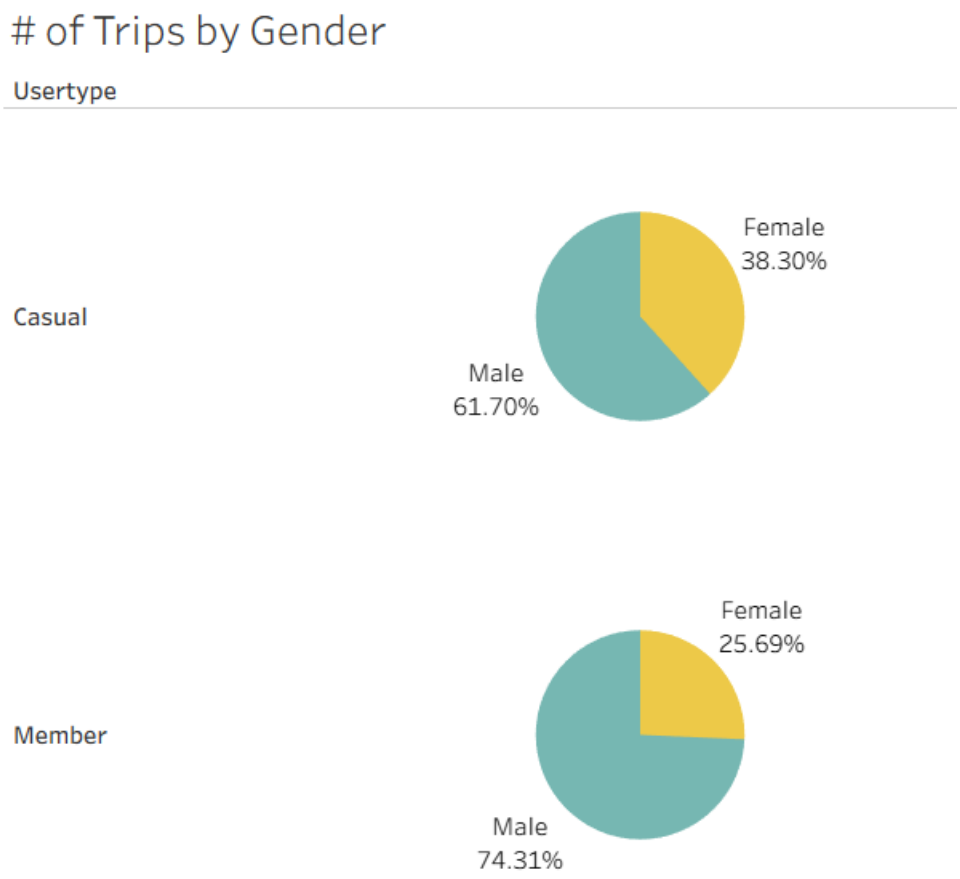

Figure 3: # of Trips by Gender- Casual rides have a larger portion of rides by female riders (38.30%) than annual members (25.69%).

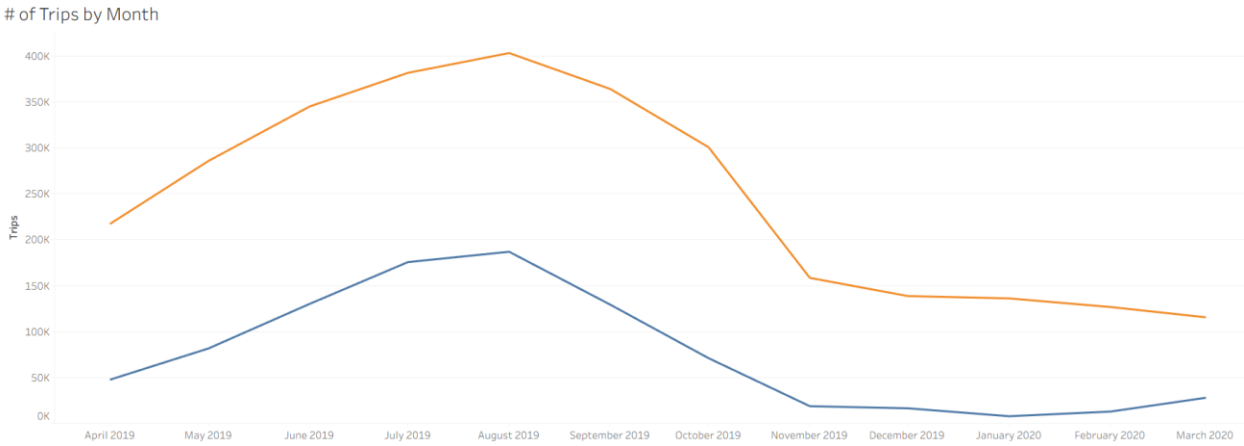

Figure 4: # of Trips by Month- Both rider types peak in August 2019, with casual ridership dropping more steeply than annual members afterwards.

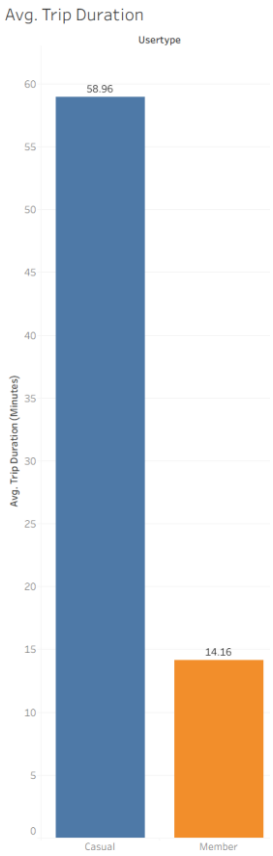

Figure 5: Avg. Trip Duration- Casual riders' average trip duration is 4 times longer than annual members.

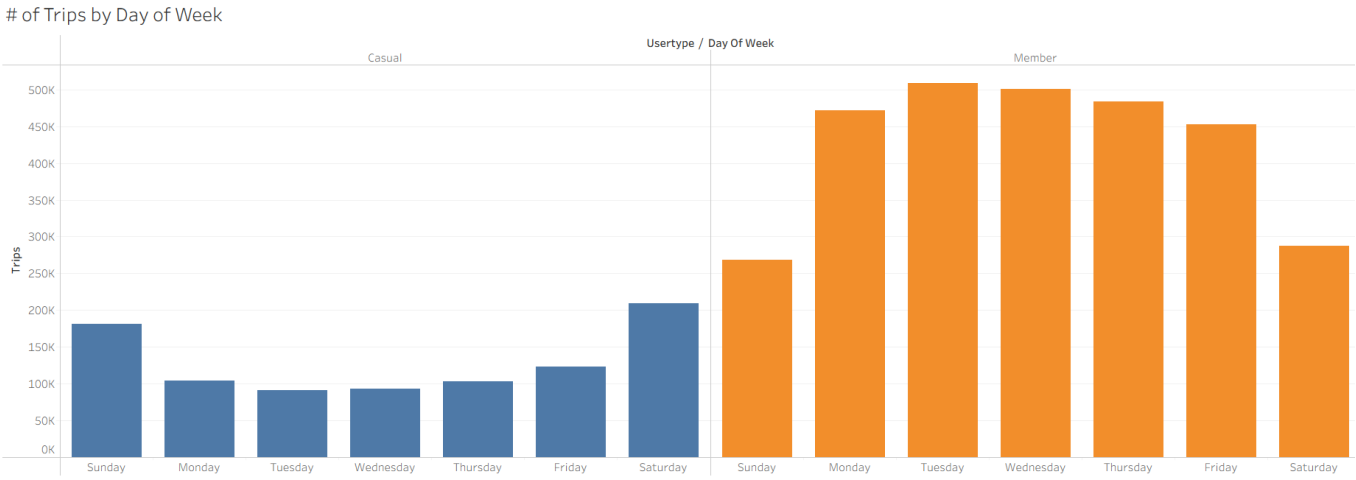

Figure 6: # of Trips by Day of Week- Casual ridership peaks on Saturday and Sunday, the days where annual member ridership is lowest, suggesting different use cases.

*Report was finished 3/28/2026. Because of dataset and the purpose of this case study a fake date is used, and relevant period information is included in analysis.